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Development and inter-laboratory evaluation of the VISAGE Enhanced Tool for Appearance and Ancestry inference from DNA

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ABSTRACT

Responding to the growing scientific and practical interest in forensic DNA phenotyping, the VISible Attributes through GEnomics (VISAGE) Consortium was founded in 2017 with the main goal of developing and validating new and reliable molecular and statistical tools to predict appearance, ancestry and age from DNA. Here, we describe the development and inter-laboratory evaluation and validation of the VISAGE Enhanced Tool for Appearance and Ancestry inference from DNA. The VISAGE Enhanced Tool for Appearance and Ancestry is the first forensic-driven genetic laboratory tool that comprises well-established markers for eye, hair and skin color with more recently discovered DNA markers for eyebrow color, freckling, hair shape and male pattern baldness and bio-geographic ancestry informative DNA markers. The bio-geographic ancestry markers include autosomal SNPs (bi- and tri-allelic SNPs), X-SNPs, Y-SNPs and autosomal Microhaplotypes. In total, primers targeting 524 SNPs (representing a 97.6% assay conversion rate) were successfully designed using AmpliSeq into a single primer pool (i.e., one multiplex assay) and sequenced with the Ion S5. In a collaborative framework, five VISAGE laboratories tested the VISAGE Enhanced Tool for Appearance and Ancestry on reproducibility, sensitivity, genotyping concordance, mixtures, species specificity and performance in relevant forensic conditions, including inhibitor-spiked, mock casework and artificially degraded samples. Based on our results, the VISAGE Enhanced Tool for Appearance and Ancestry is a robust, reproducible, and – for the large SNP number - fairly sensitive MPS assay with high concordance rates. With the VISAGE Enhanced Tool for Appearance and Ancestry introduced here, the VISAGE Consortium delivers the first single DNA-test for combined appearance prediction based on seven traits together with bio-geographic ancestry inference based on major continental regions for separated bi-parental and paternal ancestry, which represents the most comprehensive validated laboratory tool currently available for Forensic DNA Phenotyping.

1. Introduction

Forensic DNA Phenotyping (FDP) has gained attention within the forensic community from the benefits it can provide to routine forensic casework to solve criminal cases that lack known suspects. The scientific and technical advances on FDP has led to changes in forensic DNA legislations in several countries allowing FDP applications in forensic casework and other countries are allowed to practically apply FDP

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without legislative changes based on their legal situation [1]. FDP can provide new investigative leads in cases where STR-profiling fails to produce a DNA-match with known suspects included in national (or international) forensic DNA databases and/or reference profiles of case suspects [2]. In such cases, the estimation of appearance traits, biogeographical ancestry (BGA), and chronological age allows to narrow the list of putative suspects. Therefore confirming or rejecting eyewitness descriptions and providing new investigative information in cases that otherwise remain unsolved, including cold cases unsolved for years or decades.

With the advent of Massively Parallel Sequencing (MPS) technologies, the limitations on the number of DNA markers that can be simultaneously genotyped in a single assay are much less restrictive compared to non-MPS technologies used previously such as SNaPshot™ Minisequencing. Several panels previously designed to be run with (several) SNaPshot™ assays now have more reliable designs by adaptation to MPS methods. For instance, the well-established combination of tools for eye, hair and skin color prediction: IrisPlex, HIrisPlex and HIrisPlex-S [3–5], which were introduced via two SNaPshot assays for the three pigmentation traits have now been converted into a single MPS assay of different kinds, such as the HPS-NGS panel [6], the Ion AmpliSeq Phenotyping panel [4], the Ion AmpliSeq PhenoTrivium panel [7], the Ion AmpliSeq VISAGE Basic Tool for Appearance and Ancestry (VISAGE BT A&A) [8], the PowerSeq VISAGE Basic Tool for Appearance and Ancestry [9], the ForenSeq VISAGE Basic Tool for Appearance and Ancestry [10], and ForenSeq DNA Signature Prep kit B [11] via the two MPS platforms most commonly used in forensic DNA analysis, i.e., the Illumina/VEROGEN MiSeq FGx and the ThermoFisher Scientific Ion S5 instrument series. Several MPS tools have also been designed for BGA inference, both commercially available, such as the Precision ID Ancestry panel [12] and the ForenSeq DNA Signature Prep B [11] or community developed panels [7–10,13–21]. Recent developments include assays that simultaneously provide pigmentation trait prediction (based on the HIrisPlex-S and other markers) and BGA inference, e.g., the commercially available ForenSeq DNA Signature Prep B, and the AmpliSeq community panels: VISAGE Basic Tool for Appearance and Ancestry and the PhenoTrivium panel. Most of these recently introduced MPS assays were designed to allow appearance and BGA inference with separate assays, requiring multiple use of pressures evidence DNA when appearance and ancestry inference is envisioned. The few existing MPS tools that combine appearance and BGA informative DNA markers in one assay are limited in the appearance and ancestry information they provide.

Here, we overcome previous limitations in FDP tools by introducing the forensically validated VISAGE Enhanced Tool for Appearance and Ancestry inference from DNA (in the following referred to as VISAGE ET A&A), with enlarged marker sets for both purposes, appearance and ancestry prediction, and an expanded set of physical traits. The VISAGE Consortium (http://www.visage-h2020.eu/) is a Horizon 2020 EU funded project, launched in 2017, with the main goals of developing and validating robust molecular and statistical tools for appearance, BGA, and age prediction from forensic DNA samples. The VISAGE ET A&A was built from the previous VISAGE BT A&A and adds several new SNP markers for prediction of more appearance traits including eyebrow color [22], freckles [23], hair morphology [24] and male pattern baldness [25] as well as a revised marker set for BGA inference. For BGA analysis, the VISAGE ET A&A includes ancestry-informative DNA markers of different types such as Microhaplotypes, autosomal bi-allelic and tri-allelic SNPs, X-chromosomal SNPs and Y-chromosomal SNPs [26]. We present the inter-laboratory evaluation and performance assessment of the VISAGE ET A&A, as the first DNA tool of its kind that allows analysing appearance-informative markers for multiple traits including but beyond eye, hair and skin color together with different types of BGA-informative markers for bi-parental and paternal ancestry inference.

2. Materials and methods

2.1. DNA markers

Aiming to increase the number of appearance traits analysed in the VISAGE ET A&A, relative to the VISAGE BT A&A, a list of markers was selected for this design including the 41 HIrisPlex-S SNPs for eye, hair and skin color [3–5], SNPs for eyebrow colour, of which 6 do not overlap with HIrisPlex-S markers [22], SNPs for freckles of which 8 do not overlap with HIrisPlex-S markers [23], 38 SNPs for hair shape [24] and 117 SNPs for male pattern baldness [25]. In parallel, the number of BGA markers also increased, comprising 104 autosomal SNPs, 21 autosomal Microhaplotypes combining 3, 4 or 5 SNPs in close proximity, 16 X-SNPs and 87 Y-SNPs. Details on assay design are given below. A complete list of DNA markers included in the final MPS assay and their references are displayed in Supplementary Table S1 and Fig. 1.

2.2. Assay design, protocol and data analysis

The Ion AmpliSeq Designer algorithm (https://ampliseq.com/; (ThermoFisher Scientific, TFS, Waltham, MA, USA) was used to design the VISAGE ET A&A multiplex. The design took the following aspects into consideration: i. the incorporation of all markers into one primer pool to avoid multiple sample extracts and, ii. short amplicon designs to increase performance with degraded samples. The Ion AmpliSeq Designer has been utilized before in similar assays for forensic purposes, providing successful designs in identification [27,28], appearance [6–9], ancestry [7,8,15,17,19,20] and mitochondrial DNA [29] typing panels.

All DNA samples were converted into libraries using a fully automated library preparation protocol with the Precision ID DL8 Kit chemistry and the Ion Chef System (TFS, [30]). Each DL8 batch consisted of eight libraries and produced one library pool. All produced pools were quantified using the Ion Library Quantitation Kit (TFS) following the manufacturer's instructions and two pools (16 libraries in total) were combined in equimolar proportions at 30 pM (when it was not possible to reach this concentration, undiluted pools were used). Final library pools were loaded onto the Ion Chef System for template preparation with the Ion S5 Precision ID Chef & Sequencing Kit (TFS). Finally, enriched libraries were loaded automatically onto Ion 530 Chips (TFS) by the Ion Chef System and sequenced using the Ion S5 System (TFS). Raw data were aligned against the hg19 genome assembly using the TMAP aligner in the Torrent Suite (TS) software V.5.10.0 and higher (TFS). All BAM and BAI files were inspected using the Integrative Genomic Viewer (IGV, [31]). The genotypes, coverage and other metrics were extracted using the HID_SNP_Genotyper V.5.2.2 (herein, SNP Genotyper) using default parameters and InDels were called using the variant caller V.5.6.0.4 plugin. Microsoft Excel and R (https://www.r-project.org/ [32]) were used for statistical analysis and data manipulation.

2.3. Inter-laboratory validation

An extensive validation plan was designed to incorporate common developmental tests including reproducibility, sensitivity, mixed DNA typing, challenging DNA samples and specificity. Challenging samples included mock casework samples, artificially degraded DNA and inhibitor-spiked DNA. All tests were divided among VISAGE collaborating laboratories to avoid redundancy but still maintaining all replicates necessary to validate the assay. The distribution of tests per laboratory is represented in Supplementary Table S2 and each test is briefly described below.

2.3.1. Reproducibility, sensitivity and increased PCR cycles

Reproducibility and overall robustness of the assay was assessed by preparing triplicates of 2800 M control DNA (Promega, Madison, WI,

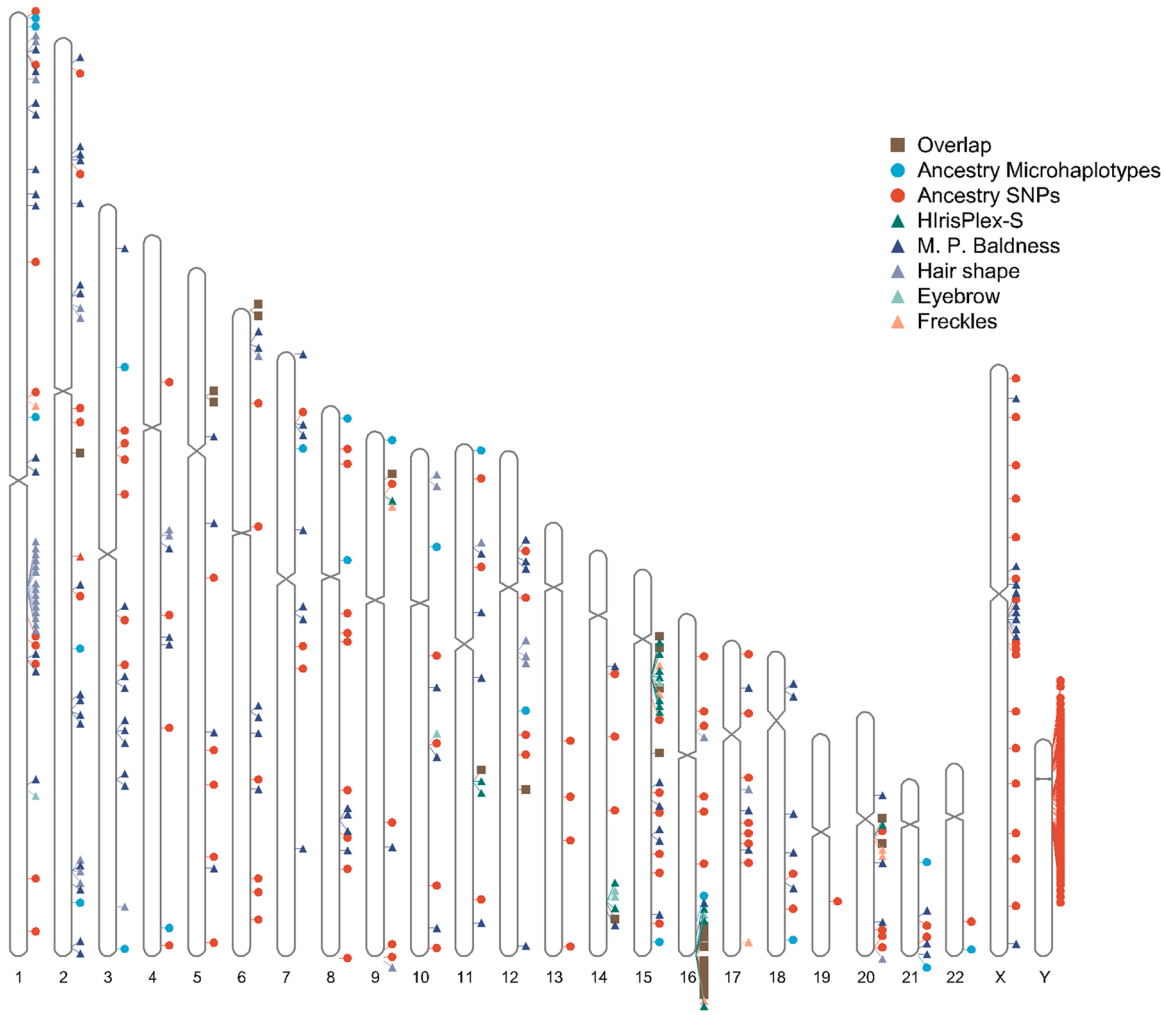


Fig. 1. Ideogram representation of the genomic location of all SNPs included in the VISAGE ET A&A assay.

USA) at three different DNA inputs (1 ng, 2 ng and 10 ng). Two laboratories participated in this test (laboratories 1 and 5) for a total of six replicates of each input. For the 1 ng replicates, we added the duplicates also prepared for the sensitivity test (2 and 10 ng: n = 6; 1 ng: n = 10). Sequencing results enabled the study of reproducibility in sequence coverage, read depth per marker and normalized read depth, to verify the assay's performance and variation with different optimum inputs. This test also allowed the estimation of important metrics including strand bias and nucleotide misincorporation rates per SNP. Finally, running optimum DNA input samples enabled the detection of underperforming SNPs and their characterization.

A sensitivity dilution series was prepared from 1 ng to 0.01 ng (1, 0.5, 0.25, 0.1, 0.05, 0.025 and 0.01 ng) of 2800 M input DNA (Promega) in duplicates by two VISAGE collaborating laboratories (laboratories 1 and 5, n = 4 per dilution). This test explored the assay's DNA input limit required to obtain a full profile and allowed identification of genotype inconsistences in the lower DNA dilutions. Furthermore, an increased PCR cycle number protocol (27 cycles instead of the standard 22) was tested, as suggested by the manufacturer for cases with low DNA quantities. The same dilution series protocol was applied for this test and performed by two different VISAGE laboratories (laboratories 2 and 4). In total, two replicates (one per laboratory) were analysed for each dilution step (n = 2).

2.3.2. *Genotyping concordance and mixed DNA samples*

Coriell cell-line DNA samples NA06994, NA07000, NA18498, NA11200 and NA07029 were used to test genotyping concordance comparisons, which comprised: i. genotyping concordance among the participating laboratories, ii. genotyping concordance between genotype calls using the VISAGE ET A&A and publicly available genotypes in the curated databases of 1000 Genomes Phase III [33] and Simons Foundation Genome Diversity Project [34], and iii. Mendelian inheritance concordance from testing the father-mother-son trio (NA06994, NA07000 and NA07029, respectively). All samples were tested at 1 ng input in three different VISAGE laboratories (laboratories 1, 2 and 3, n = 3 per Coriell sample).

Mixed DNA samples were prepared from Coriell DNAs NA07000 (European) and NA18498 (African) to explore the effect of contrasting ancestries between component of a simple 2-way mixed DNA, thus increasing the chances of obtaining different alleles. Ratios of 1:1, 1:3 and 1:9 at 1 ng input were prepared with NA18498 (male DNA sample) as the minor component. Mixture deconvolution is not a prime objective for externally visible characteristics (EVC) and/or BGA marker panels, as usually these follow STR typing methods that would identify a mixed profile. In such cases, reliance on EVC and BGA panels is not recommended since the deconvolution of profiles with SNPs is challenging and can lead to erroneous predictions of contributor genotypes. Nevertheless, as more studies are being published on mixture deconvolution using binary SNP panels [6,27,35,36], we included these analyses in the validation plan. Mixture detection and deconvolution using SNP panels has been performed by recording the imbalance of allele frequencies in different SNPs and increased heterozygosity that stems from two or

more different contributors. Three VISAGE laboratories participated in this test and produced one replicate of each mixture ratio (laboratories 1, 2 and 3, n = 3 per ratio).

2.3.3. Challenging samples

2.3.3.1. Mock samples. Mock casework samples were used to test the assay's performance with different body fluids and extraction methods. Seven GEDNAP (German DNA Profiling Group, https://www.gednap.org) proficiency testing samples (3 blood samples, 2 saliva samples and 2 semen samples) were used to mimic casework type conditions. The GEDNAP traces were sent to all VISAGE participants (laboratories 1–5, n = 5 per sample). Laboratories were asked to extract and quantify DNA following their internally validated casework methods. All GEDNAP samples were tested at 1 ng input DNA, and all quantification results and sample descriptions are detailed in Supplementary Table S3.

2.3.3.2. Inhibitor-spiked samples. Recently published studies showed MPS PCR-based methods to be more sensitive to inhibitor presence than common STR/capillary electrophoresis methods used in routine forensic profiling [8,37]. Aiming to assess the inhibition tolerance of the VISAGE ET A&A and particularly of the Precision ID DL8 Library Preparation kit, three PCR inhibitors, hematin, indigo carmine and humic acid, were used to spike 2800 M control DNA at varying concentrations. As the library preparation method is fully automated and no information on the final reaction volume is given by the manufacturer, the final inhibitor concentration cannot be calculated. Therefore, all inhibitor levels are described in total amount used: hematin varied between 8×10^{-4} - 2.5×10^{-5} μmol, humic acid varied between 1600 and 50 ng and indigo carmine between 0.16 and 0.005 μmol. Aliquots of inhibitor stocks were sent to laboratories 1–4, who were asked to prepare 2800 M replicates of 1 ng with inhibitor spikes at different concentrations (n = 2 per inhibitor concentration), two replicates of each concentration per inhibitor were prepared with the VISAGE ET A&A.

2.3.3.3. Artificially degraded samples. Artificially degraded DNA samples were used to test the assay's performance with shorter DNA fragments. Artificially degraded DNA samples were prepared from 007 control DNA (TFS) at 2 ng/μL. A sonication series (0–360 min) was prepared using an ultra-sound cleaner at 40 kHz. Degradation was verified by STR - CE typing of all samples and sonication time periods used. The AmpFlSTR NGM Select Express kit (TFS) was used to type all samples, and an overview of the results can be seen in Supplementary Fig. S1. The artificially degraded DNA samples were sent to two VISAGE collaborating laboratories (laboratories 1 and 4) who produced in total two replicates per sonication time (n = 2).

2.3.4. Species specificity

A total of 14 non-human DNA samples [38] (dog: *Canis lupus familiaris*, cattle: *Bos taurus*, sheep: *Ovis aries*, pig: *Sus scrofa*, mouse: *Mus musculus*, rat: *Rattus rattus*, goat: *Capra aegagrus hircus*, horse: *Equus caballus*, cat: *Felis catus*, donkey: *Equus asinus*, chimpanzee: *Pan troglodytes*, bonobo: *Pan paniscus*, orangutan: *Pongo sp.* and gorilla: *Gorilla gorilla*) were analysed using the VISAGE ET A&A at 1 ng input. The samples were selected based on availability, relevance in routine casework (e.g., domestic animals) and genetic similarity to human DNA (e.g., other primates). A list of all animal samples analysed is given in Supplementary Table S4.

3. Results

3.1. Assay development

A total of 537 SNPs were considered for the VISAGE ET A&A design, of which 524 were successfully included in the single MPS assay, representing an assay conversion rate of 97.6%. Fig. 1 describes the assay composition for the different appearance traits and ancestry markers (biparental and paternal). Of the 13 dropped SNPs, the majority were specific for male pattern baldness inference [25] and one BGA Microhaplotype. The final assay contained for ancestry inference: 104 autosomal SNPs (including 75 bi-allelic and 26 tri-allelic SNPs), 16 X-SNPs, 87 Y-SNPs and 21 Microhaplotypes [26]; and for appearance trait inference: 41 SNPs for eye, hair and skin color from the HIrisPlex-S system [3–5], 6 additional SNPs for eyebrow color i.e. 24 eyebrow color SNPs in total [22], 8 additional SNPs for freckles prediction i.e., 22 freckles SNPs in total [23], 38 SNPs for hair shape [24], and 107 male pattern baldness SNPs [25]. Of the 524 SNPs, a total of 26 SNPs overlapp between different categories, of which four HIrisplex-S SNPs are also used for ancestry analysis (two of these are also used for eyebrow color prediction) and one hair shape SNP also used for ancestry analysis (Supplementary Table S1 and Fig. 1).

Considering its potential applicability to forensic casework, we aimed to design an MPS assay that had an amplicon range as short as possible (an Ion AmpliSeq Designer parameter of 125–175 bp was used). As a result, the VISAGE ET A&A showed a balanced size range across all 411 amplicons (average of 116.7 ± 16.34 bp, Supplementary Fig. S2). Only four amplicons had sizes above 175 bp (rs6658216, rs142020459, rs9388490 and rs2489250), due to the presence of repetitive regions near the target SNP site that possibly prevented the design of closer primers. Earlier developmental runs to assess the overall performance of the assay included optimum input control DNA samples and a sensitivity dilution series. Further protocol optimization was found unnecessary. After preliminary evaluation of the data, the assay was distributed to five VISAGE laboratories to perform a comprehensive inter-laboratory evaluation and validation testing of the assay (see 3.2), and all data were returned and analysed by Lab 1.

3.2. Inter-laboratory assay validation

Five VISAGE Consortium laboratories participated in the evaluation and validation of the VISAGE ET A&A assay. In total, seven S5 initializations, amounting to 13 Ion 530 chips (208 samples) were run to accommodate all tests necessary for the assay validation. Aiming to streamline the validation process, different tests were distributed among the different laboratories to limit the MPS costs, while still producing all replicates necessary for comparative validation analysis (Supplementary Table S2).

3.2.1. Assay characterization and performance

Run quality was analysed using the Torrent Suite output for all runs, except for laboratory 3 data, which was not submitted (Supplementary Table S5). Run data displayed good loading values (all >78%) indicating that the majority of chip wells contained an Ion Sphere Particle (ISP). Furthermore, enrichment was always at 100%, indicating that all the ISPs presented a key signal for either real library or test fragments, indicating good performance of the Ion Chef automatic templating. Clonality represents the number of ISPs (of enriched ISPs) presenting only one template molecule and thus, only one sequencing signal. All runs showed a clonality above 50% (average 62%), in line with other AmpliSeq-based libraries. The percentage of final library represents the number of reads passing all quality filters normalized by the number of clonal reads (number of reads from an ISP with a single template molecule) and reached an average of ~77%.

3.2.1.1. Coverage. Total number of reads per reproducibility sample (replicates of 1, 2 and 10 ng) was obtained from the TS output and compared with the sum of the target reads per amplicon (Supplementary Table S6). The percentage of target reads per sample was calculated for all replicates and compared among DNA inputs. Averages of 93.32%, 93.86% and 94.32% were observed for different DNA inputs of 1, 2 and

10 ng, respectively, indicating appropriate sequencing throughput. Most of the non-targeted reads likely resulted from adapter and primer-dimer amplification. Interestingly, in terms of total number of reads (and total number of targeted reads), the 1 ng replicates showed the highest overall average (1,125,584), followed by the 2 ng (849,054) and the 10 ng (617,478) replicates (Fig. 2A). A t-test was performed comparing the total number of reads between different inputs and showed significant difference only for the 10 ng comparisons. Ion 530 chips read throughput varied from 15 to 2 million reads, when dividing these values by 16 (number of samples per chip) we obtained a theoretical interval of total reads per sample of 937,500–1250,000 reads. Only the 1 ng replicates showed values of total reads within the expected interval (Fig. 2A). Indeed, in a previous study using the same chemistry [8], we also observed the protocol to be fully optimized for 1 ng input DNA. Nevertheless, the findings of this study showed that excessive DNA input might also impair library preparation. These results were also reflected in the mean values per amplicon for the same input amounts (Supplementary Table S6), which showed 2553 mean read depth per SNP for all 1 ng replicates on average, 1934 reads for 2 ng and 1413 reads for 10 ng replicates. Read depth distribution was analysed per SNP to explore putative underperforming SNPs with systematic low coverage. Replicates of 1 ng showed 93% of the markers with an average read depth above 1000 reads and only 1.5% (8 SNPs) of the markers with mean read depth below 200 reads. Even though the percentage of markers above the 1000 read mark decreased, on average, with 2 ng (85%) and 10 ng (73%) replicates, the number of SNPs below the 200 read mark increased only slightly to 2.3% (12 SNPs) and 2.7% (14 SNPs), respectively. Linear regression analysis (Supplementary Fig. S3A) was performed to understand whether the successful amplification was related to amplicon size, but no clear relation was observed (r^2 = 0.0159). Indeed, the assay's design appeared well-balanced and the amplicon size variation was small. Normalized read depth (by total targeted reads per sample) allows the exploration of assay stability in terms of read depth distribution per SNP with varying DNA inputs. Fig. 2B shows the normalized read depth distribution of 1 ng replicates against other inputs and 2 ng vs 10 ng. We observed an overlap between all inputs, reinforcing the stability of the assay when varying input, which was confirmed by a Kolmogorov-Smirnov test (p-values > 0.05 for all input comparisons). For an ideal read distribution among amplicons, the theoretical normalized read depth would be ~0.0024 (1/512) - represented in the graph as a dashed red line. Average, in addition to median, values of normalized read depth per SNP for all replicates fell close to the ideal value of 0.0024. The number of SNPs falling under the 1st quartile (grey dashed line in Fig. 2B) were on average 17.97% for 1 ng, 17.94% for 2 ng and 18.36% for 10 ng replicates. Overall, smaller percentage values than those obtained for the previously published VISAGE BT A&A (~26%) [8] were observed with the VISAGE ET A&A.

3.2.1.2. Strand bias and nucleotide misincorporation rates. Assessment of strand bias and rate of nucleotide misincorporation per SNP was performed using all 1 ng replicates of 2800 M control DNA (n = 10). Strand bias is calculated from the ratio between forward and total number of reads and should ideally vary within the range of 45–55% for a similar number of forward and reverse reads. All replicates showed an average strand bias across all SNPs of 50.25 ± 0.24 very close to the perfect mean value. However, closer inspection of variation of strand bias per SNP showed some markers diverged from this optimum interval (Supplementary Fig. S4A). In total, 13.2% of SNPs showed average strand bias above 55% and 12.2% had averages below 45%, indicating an effect which is likely linked to amplicon and alignment issues. The rate of nucleotide misincorporation is based on the number of reads with an erroneous base at the SNP target site normalized by the total number of reads. The mean percentage of misincorporated reads per SNP (n = 10) is shown in Supplementary Fig. S4B. We observed that most markers fell below or close to an average value of 0.21%, as indicated by a median value of 0.065%. Ten SNPs had values above 1%, and of these only rs74868796 (hair shape) had a significant nucleotide misincorporation rate of 33% and was later removed from the analysis panel.

3.2.1.3. Allele read frequency balance. Allele read frequency (ARF) was calculated as the ratio between the reference allele reads over the sum of reference and alternative allele reads. Ideally, reference allele homozygotes will have values close to 1, and alternative allele homozygotes values close to 0. All reproducibility replicates (1–10 ng) and Coriell samples were used to estimate ARF balance across the ET assay marker set. In addition, all sensitivity replicates (1–0.01 ng) were analysed to

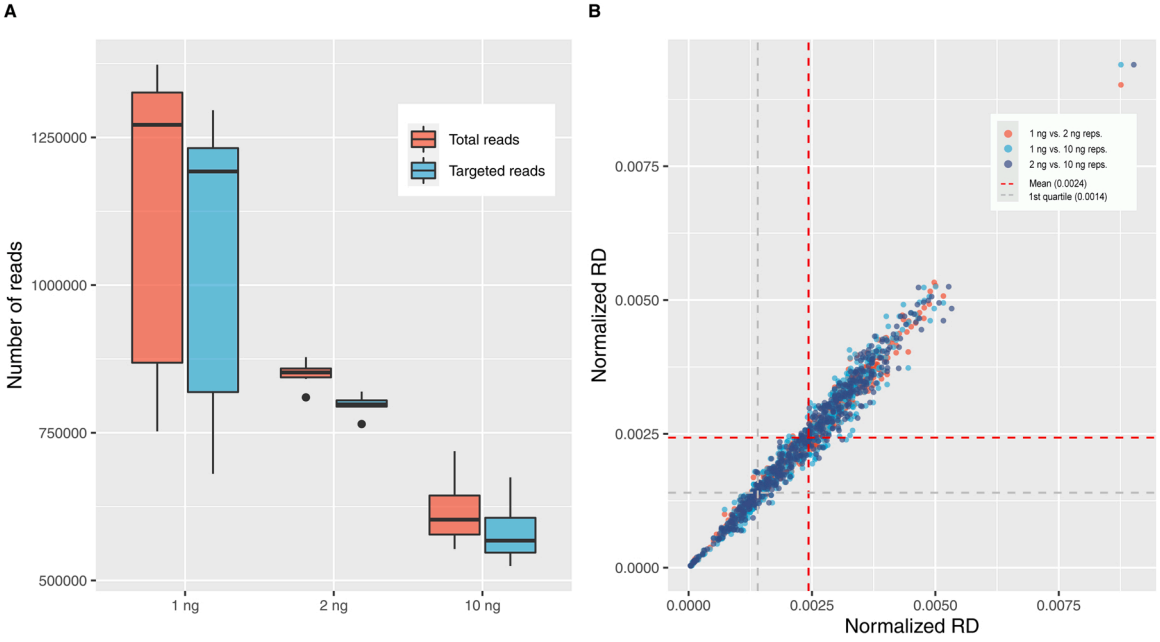


Fig. 2. A) Distribution of total reads and targeted reads for all 1 ng, 2 ng and 10 ng replicates (n = 6 for 2 and 10 ng and n = 10 for 1 ng). B) Mean normalized read depth (number of reads at one amplicon / total number of targeted reads) of 1 ng versus 2 ng (red dots), of 1 ng versus 10 ng (light blue) and of 2 ng versus 10 ng (dark purple). The red dashed line represents the mean value of 0.0024 and light grey dashed line represents the 1st quartile (0.0014).

measure variation in ARF balance with decreasing DNA input. Although heterozygotes have ideal values around 0.5, an ARF interval of 0.4–0.6 was considered as a reasonable range. Supplementary Fig. S5 outlines ARF values obtained from all reproducibility and concordance replicates. A total of 74 SNPs had at least one value outside the ideal interval for optimum input samples (≥ 1 ng). However, only 11 SNPs showed ARF values outside the expected interval in at least four different replicates (either Coriell or 2800 M at optimum input). Of these, seven needed to be removed from the prediction models due to alignment problems (e.g., highly repetitive regions; see Section No. 3.2.1.5 for detailed descriptions). The number of SNPs with values outside the ideal ARF intervals on a sample-by-sample basis were also explored. Four samples showed at least nine SNPs with altered values, two replicates of NA07029 and two replicates of NA06994. A relaxed heterozygote interval of 0.35–0.65, (see Supplementary Fig. S5), lowered the average number of underperforming SNPs from 4.7 to 2.2. In samples with less than 1 ng input, heterozygote ARF balance was maintained within the expected intervals for 0.5 and 0.25 ng; with increased variability observed in samples ≤ 0.1 ng (Supplementary Fig. S6). These results confirm the sensitivity observations described below (Section No. 3.2.2.) and are in agreement with the overall tendency of PCR-based MPS methods available for forensic DNA analysis [7–9,15,19,27,28,39].

3.2.1.4. Sequencing baseline. Non-template controls (n = 15) were used to analyse the read depth at marker-specific target sites that could potentially lead to erroneous genotype calling (Supplementary Fig. S7). Although average read depth per position reached 10.94 reads, the median read depth was considerably lower (1.93), clearly indicating an upward bias caused by certain SNPs with higher values (Supplementary Fig. S7). Overall, 134 SNPs (~26%) had an average read depth above the assay average of 10.94 reads and 100 SNPs (~19%) showed average read depth above 20 reads, which has been defined internally as a minimum read depth threshold. However, when inspecting each negative control separately, we observed an average of 13 SNPs with mean read depth above 20 reads (ranging from 26 to 4 SNPs per replicate). By calculating the read frequency considering the total number of target reads, the overall average reached 0.19% and a median of 0.048%, falling below the 2% expected for allele calling. A 2% threshold is established for detecting the presence of minor alleles, in case of mixtures or degraded samples with unbalanced alleles. In fact, only one SNP (rs17594358) showed an average read frequency above 2%, nevertheless, this raised value was caused by only one replicate and when removed from analysis, the average value was reduced to 0.063%. Overall, we observed inconsistent stochastic amplification at different target sites, which did not seem to affect allele calling in general. When analysing all replicates separately, only 30 SNPs (an average of 2 loci per replicate) showed read depth frequency above 10% - needed for allele calling in severely imbalanced heterozygotes.

3.2.1.5. Underperforming SNPs and IGV inspection. Low performance SNPs were flagged by considering systematic low sequence read depth (normalized read depth < 0.00009), strand bias (average <20% and >80%) and a high nucleotide misincorporation rate (total misincorporation > 2%). Additionally, we performed an extensive analysis of each SNP alignment for all reproducibility and concordance samples. A manual inspection of each SNP using IGV enabled detection of potential alignment issues due to the presence of highly repetitive regions, homopolymeric tracts or InDels in flanking sequence next to the target site. A list of excluded SNPs, problematic SNPs and comments on the alignment is provided in Supplementary Table S7. In total, 18 SNPs were removed (Supplementary File S1) from the analysis and 9 SNPs were flagged as low performers which require manual inspection in IGV. Supplementary Table S7 and Supplementary File S1 provide examples and analysis tips for correct sequence alignment interpretation to verify the genotype call given by TFS SNP Genotyper. When SNP Genotyper fails to provide a genotype (either because of alignment issues or low coverage), the user can manually inspect the data in IGV and compile the genotypes accordingly. In the case of low performing SNPs in non-dynamic predictive models, i.e., statistical analyses that cannot tolerate missing SNP genotypes, an option could be to input all three possible genotype possibilities and evaluate the variation of the predictions caused by these particular SNPs.

3.2.2. Sensitivity

Assay sensitivity with the standard 22-cycle PCR protocol or the 27-cycle PCR protocol was evaluated by four participating laboratories. Fig. 3A indicates the sequence read depth distribution for the 22-cycle PCR had a wider range for 0.5 and 0.1 ng replicates than the remaining inputs and with increasing dilution from 0.1 ng to 0.01 ng, a uniform decrease in read depth was observed. The 27-cycle PCR yielded decreasing read depth distribution at 1 – 0.1 ng. For the lowest dilution, however, the 27-cycle PCR provided improved sensitivity. Read depth variability between replicates of the same dilution and PCR protocol was low, even among different laboratories (Supplementary Fig. S8). Interestingly, the lower read depth values observed for the higher inputs with the 27-cycle PCR were also reflected in the number of called loci (Fig. 3B), which reached only 97.07% on average (1 – 0.1 ng replicates). Below 0.1 ng, the percentage of both allele and marker dropouts (no calls) increased, reaching similar values for both PCR protocols at 0.05 ng (allele dropouts: 2.2% and 1.3%; no calls: 1.7% and 2.1%, for 22- and 27-cycle PCRs, respectively). We also observed slightly lower read depth values for the 27-cycle PCR at 0.025 ng (allele dropouts: 6.2% and 7.6% and no calls: 8.0% and 5.8% for 22- and 27-cycle PCRs, respectively). For the lowest dilution, better performance was achieved using 27-cycle PCR, with an average of 65.7% of correct genotype calls in comparison to 58.4% for 22-cycle PCR. Even though the increase of allele dropouts was not substantially different from 0.025 ng to the lowest dilution of 0.01 ng (22-cycle PCR: 6.2–10%, 27-cycle PCR: 7.6–10.4%), the increase in locus dropouts was more marked (22-cycle PCR: 8–31.2%, 27-cycle PCR: 5.8–23.6%).

Overall, the VISAGE ET A&A MPS assay provided good overall performance down to 0.1 ng of input DNA, particularly when using a standard PCR of 22 cycles, which was also reflected by the allele read frequency results shown in Supplementary Fig. S6. When typing samples with the lowest DNA input, 27-cycle PCR is recommended, as suggested by the manufacturer. These results are in line with the VISAGE BT A&A results previously reported [8], and other PCR-based forensic MPS assays that typically show a sensitivity cut-off at around 100 pg of input DNA [7,9,10,15,19,27,28,39]. Another interesting result was the small increase in allele dropouts, drop-ins or other incorrect genotypes when compared with the increase of locus dropouts (no calls). For the purpose of this forensic DNA tool, which is the inference of appearance and ancestry from contact trace DNA, this is preferred over the risk of increased numbers of incorrect genotypes potentially leading to erroneous predictions. Nine markers showed an increased occurrence of marker and allele dropouts and should be analysed with particular caution (> 15 occurrences (35.7%) in all replicates of 22 and 27 cy; Supplementary Table S8).

3.2.3. Genotyping concordance

Genotyping concordance were evaluated using: i) inter-laboratory concordance, ii) concordance between the VISAGE ET A&A outcomes and data from the online curated databases 1000 Genomes and Simons Foundation Genome Diversity Project (SGDP) [33,34], and iii) Mendelian inheritance incompatibilities in a father-mother-son trio. Inter-laboratory concordance was 99.8% of total genotype comparisons, with 13 no calls and 3 discordances. A detailed description of discordances is given in Supplementary Table S9. Database comparison concordance was 99.3% for the 1000 Genomes Coriell genotypes and 99.8% for the SGDP Coriell genotypes. A total of 11 discordances were found between the VISAGE ET A&A genotypes and online databases

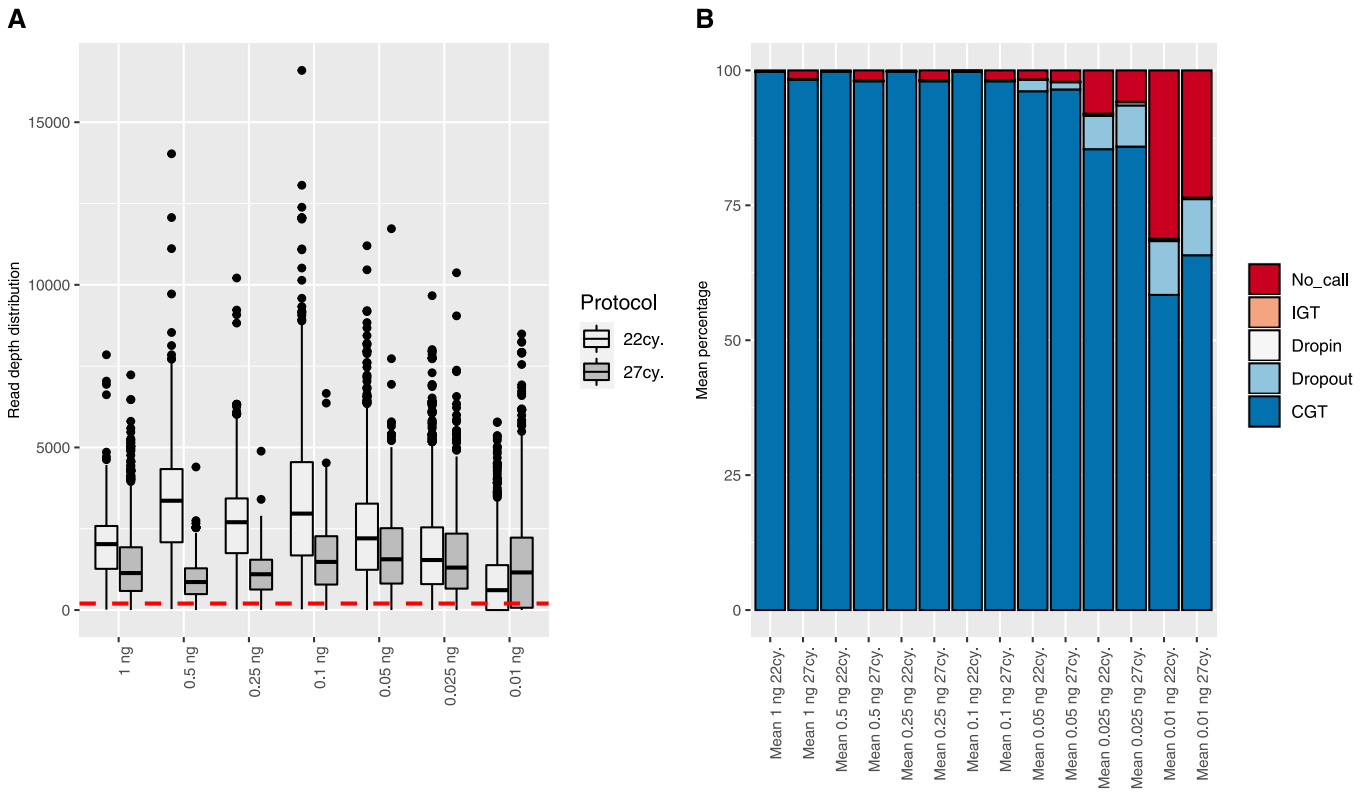


Fig. 3. A) Distribution of sequence read depth for the sensitivity dilution series replicates and different protocols (22 and 27 PCR cycles). B) Mean percentage of correct genotype calls (CGT), allele dropouts, allele dropins, incorrect genotypes (IGT) and no calls (locus dropouts) for the sensitivity dilution series and applied protocols (22 cycles and 27 cycles named PCR).

(Supplementary Table S9). Four SNPs with discordances (rs10928235, rs74333950, rs2344704 and rs71530654) were removed from the final panel due to alignment issues. A further three SNPs gave high misincorporation rates and/or high strand bias (rs745977, 8pA_rs10097211 and rs2789823) and were considered for removal. Manual inspection of the alignments using IGV helped explain certain discordances; and the relevant IGV screenshots are shown in Supplementary File S2. Supplementary Table S9 outlines possible reasons suggested from visual inspection of these data. Most discordances can be explained by misalignment due to homopolymeric tracts and/or highly repetitive regions, high strand bias and imbalanced heterozygotes. Furthermore, one InDel variant could not be called with SNP Genotyper when heterozygous genotypes were present. For the efficient analysis of such insertion/deletion variants, the Variant Caller Plugin is advised. In three cases, no reasonable explanations could be obtained from the data to explain differences to the online database genotypes listed, although we cannot discount that online data has a small proportion of incorrect calls too. No incompatibilities were detected in the genotypes of the father-mother-son trio.

3.2.4. *Mixtures*

Mixture detection using binary markers is more challenging than using multiple-allele STRs. Although the VISAGE ET A&A was not designed to focus on the deconvolution of DNA mixtures, the use of tri-allelic SNPs and Microhaplotypes provides a way to detect simple 2-contributor mixed DNA. Sequence analyses using just the SNP Genotyper software has the disadvantage that only two alleles are detected. Therefore, we tried alternative software to retrieve the putative extra SNP alleles present in 2-person mixtures as well as to analyse haplotype data in Microhaplotypes. Bi- and tri-allelic SNP data was analysed in three distinct ways to detect the presence of a mixed DNA sample: i) increased heterozygosity, ii) allele read frequency fluctuations, and iii) simultaneous presence of three alleles.

Fig. 4 shows the percentage of heterozygous and homozygous genotypes in both single source samples (used to prepare the mixed DNA samples) and the mixtures at different ratios. Furthermore, we calculated from *in-silico* simulated combinations of two single source genotypes, the percentage of expected heterozygous genotypes (expected mixture, EM). As shown in Fig. 4, the percentage of heterozygous

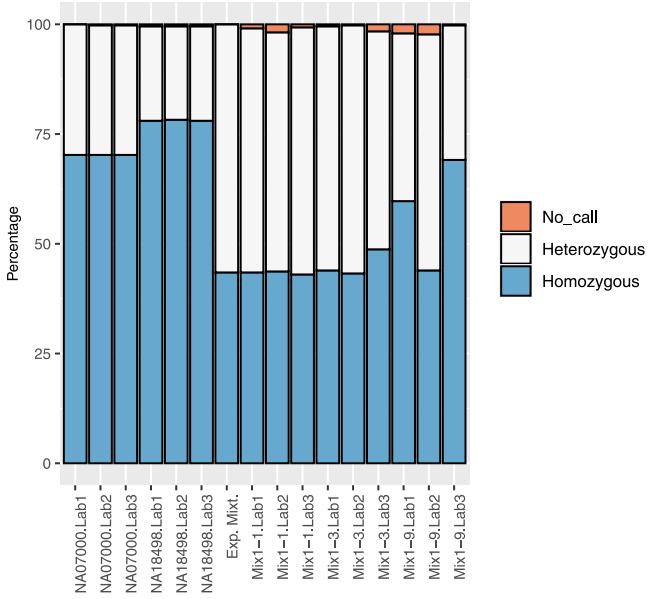


Fig. 4. Percentage of homozygous and heterozygous genotypes and no calls (locus dropouts) for all single source samples; the *in-silico* combined genotypes expected from the single source contributing samples (expected mixtures) and mixed DNA replicates analysed by participants.

genotypes in the mixed samples (average of 50%) is higher than in the single source replicates (average of 25.5%). Indeed, 1:1 and 1:3 mixture replicates showed a percentage of heterozygous values close to the theoretical expected 56.5% from EM calculations. In the more diluted minor component 1:9 mixtures, the percentage of heterozygotes generally decreased, except for one replicate staying at 53.7% heterozygous genotype calls. These results are expected since the cut-off for allele calling in SNP Genotyper is a 0.1 ARF value, as previously described [8]. Nevertheless, previous studies suggest it is viable to apply a lower threshold for allele calling to detect low level minor contributors [19,27,40].

The fluctuations in allele read frequency can also be used for mixture detection and has formed the basis for mixture deconvolution of binary markers [6,35]. A single source individual is expected to present > 90% for homozygous alleles and ~50% for heterozygous alleles thus, all variation observed outside these expected intervals for a numerous set of markers is indicative of a mixed profile. SNP Genotyper lists the major allele frequency per marker (MAF), which is flagged when a marker falls outside the expected default intervals (95–100% for homozygotes and 35–65% for heterozygotes). The number of markers falling outside the expected intervals we set for ET A&A (>90% for homozygotes and 40–60% for heterozygotes) was plotted per replicate of single source and mixed sample (Supplementary Fig. S9). Even though some markers showed MAF values outside of these intervals in the single source replicates (an average 6.6 markers across all replicates, excluding Y markers in the female sample), the values significantly increased when analysing mixed DNA samples (t-test p-values < 0.05), with average values of 177.7, 162.7 and 87.3 markers for 1:1, 1:3 and 1:9 replicates, respectively. Once again, the number of markers outside of the MAF intervals is smaller for the 1:9 ratio replicates but were still significantly different from the single source values (t-test p-value < 0.05). Only one Y-chromosome marker failed to amplify in two replicates of 1:9 mixture ratios (M123) that led to a MAF of zero. The remaining 86 Y-SNPs amplified successfully. The variation of lower read frequencies within the interval $0.02 < x < 0.4$ (Supplementary Fig. S10) was plotted to observe if the read frequency matched the expected theoretical values for the minor contributors for mixture ratios 1:3 and 1:9 for all laboratories. Depending on how diluted the minor contributor was (i.e., 1:3 vs 1:9), the expected theoretical values for contrasting homozygous genotypes and one contrasting allele varied from 0.25 and 0.125 for 1:3 and 0.1 and 0.05 for 1:9 mixture ratios, respectively. Interestingly, apart from laboratory 2 replicates, the base frequency fluctuations remained close to theoretical values. The laboratory 2 results could be explained by flawed quantitation. The absence of base read frequencies around 0.125 for the 1:3 mixture laboratory 2 replicate indicates this is the likely explanation.

The analysis of third alleles had to be performed manually by calling all alleles showing frequencies between the $0.02 < x < 0.4$ interval. Supplementary Table S10 describes all markers with three allele calls that unequivocally show presence of a different contributor. When compared with the *in-silico* expected mixture calls (Art. Mixt), all the 1:1 replicates showed the expected three allele calls in all 11 markers. However, laboratory 3 replicates also showed an additional extra allele call at rs6504633, which was previously flagged as a SNP with high misincorporation (>2%). For the 1:3 replicates, all SNPs with expected three alleles were correctly called, but artifact alleles (for all replicates) were also observed at rs2789823, similarly previously flagged for high misincorporation (>2%). Therefore, particular care should be taken when analyzing mixed DNA samples, particularly when markers have been characterized with high misincorporation rates, as these loci have a risk of erroneous allele calling. Careful visual inspection of the sequence data using IGV is recommended. Replicates of 1:9 showed not only the erroneous third allele calling in rs6504633 and rs2789823, but also 'third allele dropout' at rs1398461 and rs393953. The Microhaplotype data was analysed using a new plugin developed by TFS named Microhaplotyper. This java-based software outputs a table with the marker list, phased haplotypes and read depth data. A summary of the results for laboratory 1 replicates is given in Supplementary Table S11 and Supplementary Fig. S11. All minor alleles were successfully identified by the plugin and showed decreasing ARF values with increasing dilution (1:1–1:9), as expected.

3.2.5. Challenging samples

Challenging samples were analysed to gauge ET A&A assay performance with mock casework samples from different biological tissues and using different extraction methods established in the participating laboratories. Furthermore, inhibitor spiked samples and artificially degraded samples were included to assess the assay's tolerance and performance with shorter input DNA fragments.

3.2.5.1. GEDNAP mock samples. Seven GEDNAP samples were shared amongst participating laboratories for DNA extraction and quantification according to each participant's in-house validated protocols. Different body fluids including blood, semen and saliva were used to test the ET A&A assay performance with different extraction protocols and quantification methods. A list of the GEDNAP samples used and concentrations obtained per laboratory is outlined in Supplementary Table S3. Two replicates from laboratory 3 of samples 42S3 and 53S1 were removed from the analysis due to several loci dropout and possible contamination. All remaining genotypes were compared between replicates and on average, 99.72% of genotypes were concordant. Three samples had 3–5 discordant genotypes and no calls (49S2, 49S4 and 42S3), however, 44S3 gave 12 discordant genotypes and 7 no calls (locus dropouts). One explanation could be obtained from the quantification results of 44S3 (Supplementary Table S3). As 44S3 was a saliva sample, with the lowest DNA amount of all samples (an average 1.3 ng/μL for all participating laboratories). Two samples (45S2 and 53S1) had fully concordant results among all replicates, both also showed high DNA concentrations after extraction (averages of 20.2 and 9.1 ng/μL across laboratories). 45S2 and 49S4 were semen samples, with the highest overall quantitation results (Supplementary Table S3); while 53S1 was derived from blood on a stone. These results emphasise the applicability of the VISAGE ET A&A assay with DNA samples from different biological tissues and adaptability to each laboratory's in-house workflows.

3.2.5.2. Inhibited samples. To assess the VISAGE ET A&A assay's inhibitor tolerance, control DNA 2800 M was spiked with PCR inhibitors: hematin, humic acid and indigo carmine at varying concentrations, and with libraries prepared automatically by the Ion Chef, the final volume of the initial PCR was not calculated. Therefore, inhibitors were described in final total amounts, not final concentration per reaction. Inhibitor tolerance reached up to 4×10^{-4} μmol hematin, 200 ng of humic acid and 0.02 μmol of indigo. These results confirm those obtained in a previous study using similar AmpliSeq-based chemistry [8]. Both read depth and percentage of correct genotype calls were significantly reduced (Supplementary Fig. S12). PCR-based MPS assays are known to be more sensitive to inhibitors than conventionally applied PCR-based capillary electrophoresis typing methods [37]. In general, these results reflect many years of experience and development of forensic kit enzymes and buffers, used in the previous assays and highlights a need to improve the current MPS PCR chemistry specifically for forensic purposes.

3.2.5.3. Artificially degraded samples. The sonication time series of 007 control DNA used as artificially degraded DNA samples was compared with a routine STR-CE typing method (AmpFlSTR™ NGM SElect™ Express Kit, TFS). Supplementary Fig. S13 outlines the results obtained for both methods as percentage of correct calls, and rates of allele and locus dropout (no call). Even though a similar pattern is discernible in both methods, analysis of the sample with the longest sonication period

indicated the ET A&A assay produced 75.4% correct genotypes (average across replicates), whereas the NGM Se Ex only achieved 17.6%. Read depth distribution for all replicates reflects the genotype results obtained, showing a stable distribution up to 300 min sonication time and a decrease for the 360 min replicates. Close inspection of the CE electropherograms (Supplementary Fig. S1) showed a stable RFU scale for 0–180 min and a gradual decrease of signal for the longer sonication time replicates, particularly for the longer STR amplicons such as SE33, confirming progressive target DNA degradation. As previously described, the VISAGE ET A&A has an average amplicon size of 166.67 ± 16.4 bp (median of 121 bp), so only 16 amplicons are longer than 130 bp. Despite showing low read depth and locus dropouts in 3 of the largest amplicons in the assay (rs6658216, rs2489250 and rs9388490), no correlation was detected between amplicon size and lower read depth (Supplementary Fig. S3B).

3.2.6. Species specificity

Notably, even though amplification at the SNP target sites was only observed for Primates (Supplementary Fig. S14), the read depth distribution in the remaining species showed several outliers with high read depth (reaching up to 59,219 reads for Sus scrofa / pig sample). Such outliers may be due to PCR over-competition, combined with several failed markers, causing over amplification of certain markers. However, these outliers produce an increased Y-scale, which impairs the visualization of other samples. Apart from the non-human Primate samples, the average percentage of called loci was 1.32% (with an average read depth ranging from 19.97 to 197.44) - close to the negative control value (0.95%). In contrast, the non-human Primate samples showed an average 76.19% called loci (average read depth ranging from 1433.79 to 1843.75 reads), reaching 87% for the Bonobo. These results can be explained by the genetic similarity amongst Primates including humans; although these non-human primate species are unlikely to contaminate forensic samples in most cases. Similar results were observed in the previous evaluations of the VISAGE BT A&A assay [8], where amplification outside humans was only detected in non-human Primate samples too.

4. Discussion

The increasing interest in alternative DNA-based methods to provide investigative leads when standard STR-profiling is unsuccessful is evident in the current scientific literature and in law enforcement and social policy discussions [1,2]. Here, we present the development and inter-laboratory technical evaluation of the VISAGE Enhanced Tool for Appearance and Ancestry inference from DNA. To the best of our knowledge, this is the first forensic molecular genetic tool that integrates several appearance traits (including but beyond eye, hair, and skin color) as well as biparental and paternal bio-geographic ancestry, all in a single MPS assay using more than 500 SNPs. Overall, the assay produced highly reproducible results amongst participating laboratories and was stable when evaluating varied DNA input levels. As a PCR-based AmpliSeq assay, the VISAGE ET A&A provides comparable performance characteristics to the previously published assessments of the VISAGE BT A&A assay [8–10], which, however includes less appearance traits and less SNPs in total. Obtaining similar performance while combining nearly 3.4 times more SNPs into the single multiplex assay (153 in BT versus 524 SNPs in ET) reflects a remarkable achievement. Indeed, the run configuration recommendations previously noted for the more limited VISAGE BT A&A remain at 16 samples per 530 Ion Chip also with the VISAGE ET A&A despite the 3.4 times increase of included SNPs. Although displaying a good overall sequencing performance, some SNPs had low sequencing quality data due to flanking regions that were challenging for primer design. Therefore, we compile details of these under-performing SNPs and some recommendations for their sequence analysis using IGV.

Importantly, the VISAGE ET A&A assay maintained the same sensitivity threshold as the VISAGE BT A&A assay (~100 pg), despite the 3.4 times increase of SNPs included. The 100 pg sensitivity threshold (approximately 14 cells) to obtain full profiles is a finding that matches those of several other PCR-based forensic tools [7–10,19,27,28] and might reflect the overall limitation of PCR-based MPS library preparation reagents. One relevant finding from the VISAGE ET A&A evaluations is the increase of locus dropouts (no call) with decreasing DNA input, rather than an increase of incorrect genotypes caused by allele dropouts or dropins, as may be expected. Since the final application of this tool is the estimation of appearance traits and BGA, this effect minimizes the input of erroneous genotype data in the final appearance trait and BGA prediction, and thus minimizes wrong prediction outcomes. However, the direct effect of DNA input variation on the final prediction values is beyond the scope of this study and is the subject of a parallel study (in preparation) on the implementation of the VISAGE ET A&A assay.

High genotype concordance values (99.3% and 99.8%) were obtained when comparing genotypes produced with the VISAGE ET A&A with those of online databases, underlining the reliability of the assay. Most of the SNPs that showed discordant results also suffered from ambiguous alignments and/or high misincorporation rates - so were flagged as problematic. Although the standard procedure of forensic DNA profiling (STR-CE analysis) recognizes a mixture, the ability to detect mixtures with SNPs is currently subject to a range of research initiatives [6,27,35,36]. In principle, the addition of tri-allelic SNPs and Microhaplotypes clearly aids mixture detection and offers the possibility of mixture deconvolution going forward. As a proof-of-concept, we analysed mixtures at 1:1, 1:3 and 1:9 ratios and it was a viable approach to identify the presence of a mixture, either by allele read frequency fluctuations, increased heterozygosity, the presence of three alleles and analysis of Microhaplotype data. Notably, higher numbers of alleles present in Microhaplotypes provided more straightforward mixture deconvolution (Supplementary Table S11) than the VISAGE BT A&A assay [8–10].

Of the challenging samples used to test the VISAGE ET A&A performance, GEDNAP samples comprising different biological materials and mimicking casework conditions produced an overall genotyping concordance above 99% amongst the five participating laboratories. Inhibitor-spiked samples showed similar results as those previously published for BT [8,10] as well as other PCR-based MPS methods [37]. Recent studies indicate PCR-based MPS assays are less tolerant of inhibitors than standard STR-CE methods. Inhibitor concentrations, to which these new MPS methods are sensitive, might not be detectable with common real-time quantitative PCR assays, as previously discussed [10]. Therefore, the processing of inhibited samples and subsequent impairment of sequencing results might proceed undetected. These results not only highlight an improvement of MPS chemistry's tolerance to inhibitors, but also require the development of more sensitive real-time qPCR assays that lead to reliable inhibitor detection at these concentration levels. Finally, the artificially degraded samples produced full profiles down to 300 min sonication, while leading to partial STR profiles from standard forensic DNA profiling. The VISAGE ET A&A correctly called 75.4% (395 markers) of the profile at 360 min of sonication, while parallel STR-CE analyses only gave three correctly typed markers (17.6%). Such results underline the relevance of SNPs and smaller amplicon designs for forensic purposes. Overall, the assay gave highly concordant results among participating laboratories and worked successfully in different laboratory workflows (DNA extraction and quantification) applied to samples with varied biological origins or which were highly degraded.

In conclusion, the VISAGE ET A&A assay has proved to be a reliable and resourceful tool for Forensic DNA Phenotyping applications, providing prediction of eye, hair, skin and eyebrow color, freckles, hair shape, male pattern baldness, in addition to maternal and paternal biogeographic ancestry. Compared with the previous VISAGE BT A&A, the VISAGE ET A&A provides appearance information for four additional

appearance traits and autosomal SNP-based BGA analysis has been extended with X-SNPs and Y-SNPs, and Microhaplotypes bolstered by tri-allelic SNPs. The latter has enhanced the deconvolution of simple mixtures, which has been traditionally hampered by reliance on MPS sequencing of binary markers alone. We invite the industry to further develop and optimize the VISAGE ET A&A to make the tool accessible to a broader audience. Currently, use of PCR-based library preparation methods for MPS sequencing limits the further increase of marker numbers that can be combined into a single test while maintaining sensitivity and cost-efficiency. Recent advances in hybridization capture based MPS methods provide alternative approaches for enlarging the number of simultaneously-analysed markers and increased sensitivity levels, albeit at higher analytical costs [41–43]. Therefore, PCR-based tests, exemplified by the VISAGE ET A&A assay will continue to provide cost-effective and reliable genotyping tools for the generation of investigative leads from combined analyses of appearance, ancestry from contact forensic trace DNA and can be combined with separate assays for age prediction such as the VISAGE BT and ET for Age [44–46]. Continuous research in advancing our knowledge on appearance genetics will further increase the number of physical traits that will become predictable from DNA, which – if combined in a single assay as especially desired in forensic applications - will increase the total number of targeted SNPs to the thousands. Hence, MPS methods to simultaneously analyse thousands of SNPs in a highly sensitive, reliable and cost-effective way suitable for analysing forensic DNA will be needed to further advance FDP tools in the future.

CRediT authorship contribution statement

C. Xavier: Conceptualization, Methodology, Investigation, Formal analysis, Visualization, Writing – original draft. **M. de la Puente:** Methodology, Investigation, Formal analysis, Writing – review & editing. **A. Mosquera-Miguel:** Investigation. **A. Freire-Aradas:** Investigation. **V. Kalamara:** Investigation. **A. Ralf:** Investigation. **A. Revoir:** Investigation. **T.E. Gross:** Investigation. **P.M. Schneider:** Investigation, Validation, Writing – review & editing. **C. Ames:** Investigation, Validation, Writing – review & editing. **C. Hohoff:** Resources. **C. Phillips:** Investigation, Validation, Writing – review & editing. **M. Kayser:** Investigation, Validation, Writing – review & editing. **W. Parson:** Conceptualization, Validation, Supervision, Funding acquisition, Writing – review & editing.

Conflict of interest statement

The authors declare no conflicts of interest.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.fsigen.2022.102779.

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